Mathematics for Machine Learning

MAI391

Huynh Minh Khang

Contents

1. Analytic Geometry

1.1 Norm

Definition 1.1.1

A norm on V is a non-negative function

$$\|.\| : V \to \mathbb{R} \tag{1.1}$$

such that $\forall k \in \mathbb{R}$ and $x, y \in V$:

- *Absolute homogeneous*: $\|kx\| = |k| \ \|x\|$
- **Triangle inequality**: $\|x + y\| \leq \|x\| + \|y\|$
- **Positive definite**: $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = \vec{0}$

♣

Definition 1.1.2

Let p be as positive number. The l_p-norm $\|.\|_p$ in $\mathbb{R}^n$:

$$\|x\|_p = \sqrt[p]{\sum_{i=1}^{n} |x_i|^p} \quad \text{for any } x = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \in \mathbb{R}^n \tag{1.2}$$

♣

Example 1.1.3

- $p = 1 \to$ Manhattan norm l_1:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i| \tag{1.3}$$

- $p = 2 \to$ Euclidean norm l_1:

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} |x_i|^2} \tag{1.4}$$

- $p = \infty \to$ maximum norm l_∞:

$$\|x\|_\infty = \max\{|x_1|, ..., |x_n|\} \tag{1.5}$$

◊

Definition 1.1.4

Euclidean norm on $\mathbb{R}^{m\times n}$:

$$\|A\| = \left(\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2\right)^{\frac{1}{2}} \tag{1.6}$$

♣

Definition 1.1.5

Spectral norm on $\mathbb{R}^{m\times n}$:

$$\|A\|_2 = \max_{x\in\mathbb{R}^n,\ \|x\|_2\ =1} \|Ax\|_2 \tag{1.7}$$

♣

1.2 Inner Product

Definition 1.2.1

A linear mapping L on V is a function $L : V \to \mathbb{R}$ that satisfies:

- $L(kx) = kL(x)$
- $L(x + y) = L(x) + L(y)$

$\forall x \in V$ and $k \in \mathbb{R}$

♣

Definition 1.2.2

A bilinear mapping or bilinear form B on V is a function $B : V \times V \to \mathbb{R}$ that is **linear in each argument separately**:

- $B(kx, z) = kB(x, z)$
- $B(x + y, z) = B(x, z) + B(y, z)$
- $B(x, kz) = kB(x, z)$
- $B(x, y + z) = B(x, y) + B(x, z)$

$\forall x, y, z \in V$ and $k \in \mathbb{R}$

♣

Problem 1.2.3

Show that the mapping $B : R^2 \times R^2 \to R$

$$B(x, y) = x_1 y_1 - 2x_1 y_2 + 3x_2 y_2 \tag{1.8}$$

is a bilinear mapping on R^2.

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Solution.

$$\begin{aligned}
B(kx, z) &= kx_1 z_1 - 2kx_1 z_2 + 3kx_2 z_2 = k(x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) = kB(x, z) \\
B(x + y, z) &= (x_1 + y_1) z_1 - 2(x_1 + y_1) z_2 + 3(x_2 + y_2) z_2 \\
&= (x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) + (y_1 z_1 - 2y_1 z_2 + 3y_2 z_2) \\
&= B(x, z) + B(y, z) \\
B(x, kz) &= kx_1 z_1 - 2kx_1 z_2 + 3kx_2 z_2 = k(x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) = kB(x, z) \\
B(x, y + z) &= x_1 (y_1 + z_1) - 2x_1 (y_2 + z_2) + 3x_2 (y_2 + z_2) \\
&= (x_1 y_1 - 2x_1 y_2 + 3x_2 y_2) + (x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) \\
&= B(x, y) + B(x, z)
\end{aligned} \tag{1.9}$$

$\Rightarrow$ B is linear mapping □

Definition 1.2.4

A bilinear mapping $B : V \times V \to R$ is called

- **symmetric** if $B(x, y) = B(y, x)$
- **positive definite** if $B(x, x) > 0$ for any $V \ni x \neq \vec{0}$

♣

Definition 1.2.5

A positive definite, symmetric bilinear mapping $B : V \times V \to R$ is called an **inner product**. We typically write $\langle x, y \rangle$ instead of $B(x, y)$ for inner product

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Definition 1.2.6

The pair $(V, \langle ., . \rangle)$ is called **inner product space**

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Definition 1.2.7

The **dot product** or **scalar product** on $\mathbb{R}^n$ is given by

$$x^T y = \sum_{i=1}^{n} x_i y_i \tag{1.10}$$

♣

Property 1.2.8

Dot product is inner product, and $(\mathbb{R}^n, \langle .,. \rangle)$ is **Euclidean vector space**

♠

Problem 1.2.9

Consider $V = \mathbb{R}^2$. Define

$$\langle x, y \rangle = 2x_1y_1 + x_1y_2 + x_2y_1 + 2x_2y_2 \tag{1.11}$$

Show that $\langle .,. \rangle$ is an inner product in $\mathbb{R}^2$

◇

Solution.

- $\langle .,. \rangle$ is a bilinear mapping *(The solution is similar to Problem 1.2.3)*

- $\langle .,. \rangle$ is symmetric

$$\begin{aligned} \langle x, y \rangle &= 2x_1y_1 + x_1y_2 + x_2y_1 + 2x_2y_2 \\ &= 2y_1x_1 + y_1x_2 + y_2x_1 + 2x_2y_2 = B(y, x) \end{aligned} \tag{1.12}$$

- $\langle .,. \rangle$ is positive definite

$$\begin{aligned} \langle x, x \rangle &= 2x_1^2 + 2x_2^2 + 2x_1x_2 \\ &= (x_1 + x_2)^2 + x_1^2 + x_2^2 \geq 0 \end{aligned} \tag{1.13}$$

The equality occur only when $x_1 = 0, x_2 = 0, x_1 + x_2 = 0$ which is $x = \vec{0}$
$\Rightarrow \langle .,. \rangle > 0$ for any $\mathbb{R}^2 \ni x \neq \vec{0}$, or $\langle .,. \rangle$ is positive definite

So $\langle .,. \rangle$ satisfies all properties to be an inner product □

Definition 1.2.10

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is

- **positive definite** if

$$x^T A x > 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \tag{1.14}$$

- **positive semidefinite** if

$$x^T A x \geq 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \tag{1.15}$$

♣

Problem 1.2.11

Consider the matrices

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \tag{1.16}$$

Show that A is positive definite but B is not

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Solution.

$$\begin{aligned} x^T A x &= [x_1 \;\; x_2] \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= 2x_1^2 + 2x_1 x_2 + 3x_2^2 \\ &> 0 \;\; \text{for all non-zero vectors } x \in \mathbb{R}^n \end{aligned} \tag{1.17}$$

(the proof for why > 0 is similar to Problem 1.2.9)
Also A is symmetric so A is positive definite

$$\begin{aligned} x^T B x &= [x_1 \;\; x_2] \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1^2 + 4x_1 x_2 + 3x_2^2 \end{aligned} \tag{1.18}$$

Take $x = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow x^T B x = -1$
So B is not positive definite and not positive semidefinite □

Note 1.2.12

When multiply matrices of form $x^T Ax$ *(like in Problem 1.2.11)*, the coefficient of

$$x_i x_j = \begin{cases} a_{ij} \text{ , if } i = j \\ a_{ij} + a_{ji} \text{ , if } i \neq j \end{cases} \tag{1.19}$$

1.3 Lengths and Distances

Definition 1.3.1

The **norm** is $\|x\| := \sqrt{\langle x,\, x\rangle}$

Definition 1.3.2

The **distance between x and y** is $d(x, y) := \|x - y\| = \sqrt{\langle x - y,\, x - y\rangle}$

Theorem 1.3.3

Cauchy-Schwarz inequality

$$|\langle x, y\rangle| \leq \|x\| \, \|y\| \tag{1.20}$$

Problem 1.3.4

Let the inner product $\langle ., .\rangle$ be defined on $\mathbb{R}^n$ by

$$\langle x, y\rangle = x_1 y_1 - x_1 y_2 - x_2 y_1 + 3x_2 y_2 \tag{1.21}$$

Find the distance between $u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Solution.

$$u - v = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \tag{1.22}$$

$$\begin{aligned} d(u,v) = \|u-v\| &= \sqrt{\langle u-v,\ \ u-v\rangle} \\ &= \sqrt{2\cdot 2 - 2\cdot(-1) - (-1)\cdot 2 + 3\cdot(-1)\cdot(-1)} \\ &= \sqrt{11} \end{aligned} \tag{1.23}$$

□

1.4 Angles and Orthogonality

Definition 1.4.1

The **angle** between vector x and y is the number $\theta \in [0, \pi]$ defined by

$$\cos\theta = \frac{\langle x, y\rangle}{\|x\|\ \|y\|} \tag{1.24}$$

♣

Problem 1.4.2

Define the inner product $\langle ., .\rangle$

$$\langle x, y\rangle = x\cdot y = x_1y_1 + x_2y_2\ (\textit{dot product}) \tag{1.25}$$

Find the angle between $x = \begin{bmatrix}1\\0\end{bmatrix}$ and $y = \begin{bmatrix}-\frac{1}{2}\\ \frac{\sqrt{3}}{2}\end{bmatrix}$

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Solution.

$$\langle x, y\rangle = 1\cdot\left(-\frac{1}{2}\right) + 0\cdot\frac{\sqrt{3}}{2} = -\frac{1}{2} \tag{1.26}$$

$$\begin{gathered} \|x\| = \sqrt{1\cdot 1 + 0\cdot 0} = 1 \\ \|y\| = \sqrt{-\frac{1}{2}\cdot -\frac{1}{2} + \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}} = 1 \end{gathered} \tag{1.27}$$

$$\cos\theta = \frac{\langle x, y\rangle}{\|x\|\ \|y\|} = \frac{-\frac{1}{2}}{1\cdot 1} = -\frac{1}{2} \tag{1.28}$$

$$\Rightarrow \theta = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3} \tag{1.29}$$

□

Definition 1.4.3

Two vectors x and y are **orthogonal** (or $\boldsymbol{x} \perp \boldsymbol{y}$) $\Leftrightarrow \langle x, y \rangle = 0$

♣

Definition 1.4.4

If $\boldsymbol{x} \perp \boldsymbol{y}$ and $\|x\| = \|y\| = 1$ *(x and y are unit vectors)* then x and y are **orthonormal**

♣

Definition 1.4.5

A square matrix $A \in \mathbb{R}^{n \times n}$ is an **orthogonal matrix** if and only if

$$AA^T = A^T A = I_n \tag{1.30}$$

♣

Property 1.4.6

If $A \in \mathbb{R}^{n \times n}$ is orthogonal then

- $A^{-1} = A^T$
- A preserve the length of any vector $x \in \mathbb{R}^n$

$$\|Ax\| = \|x\| \tag{1.31}$$

- A preserve the angle between two vectors x and y in $\mathbb{R}^n$

$$\text{angle}(Ax, Ay) = \text{angle}(x, y) \tag{1.32}$$

♠

Note 1.4.7

For square matrix $A \in \mathbb{R}^{n \times n}$, A^{-1} is defined as matrix with

$$AA^{-1} = A^{-1}A = I_n \tag{1.33}$$

So if A is orthogonal, $A^T = A^{-1}$

◊

1.5 Orthonormal Basis

Definition 1.5.1

Given a vector space V, a set of vectors $\{v_1, ..., v_n\}$ is **linear independent** if the **only solution** to equation

$$a_1 v_1 + ... + a_n v_n = 0 \quad (1.34)$$

is $a_1 = ... = a_n = 0$

♣

Definition 1.5.2

Given a vector space V, a set of vectors $B = \{b_1, ..., b_n\}$ is basis if:

- **Every subset** of B is **linear independent**
- For every $x \in V$ there exist $\{a_1, ..., a_n\}$ such that

$$a_1 b_1 + ... + a_n b_n = x \quad (1.35)$$

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Example 1.5.3

Most common basis for $\mathbb{R}^2$ is $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

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Definition 1.5.4

Consider an dimensional inner product vector space $(V, \langle .,. \rangle)$ and a basis $B = \{b_1, ..., b_n\}$ of V. B is **orthonormal basis** if

$$\langle b_i, b_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (1.36)$$

♣

Theorem 1.5.5

Let $B = \{b_1, ..., b_n\}$ be an orthonormal basis of an inner product space V. If x is any vector in V, then

$$x = \langle x, b_1 \rangle b_1 + ... + \langle x, b_n \rangle b_n \quad (1.37)$$

♡

Problem 1.5.6

Show that $B = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ is an orthonormal basis of $\mathbb{R}^n$ with inner product

$$\langle x, y \rangle = x^T A y, \text{ where } A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \tag{1.38}$$

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Solution.

$$\begin{aligned} \langle b_1, b_1 \rangle &= \langle \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rangle \\ &= [1 \ \ -1] A \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= 1 \end{aligned} \tag{1.39}$$

$$\begin{aligned} \langle b_1, b_2 \rangle &= \langle \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rangle \\ &= [1 \ \ -1] A \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= 0 \end{aligned} \tag{1.40}$$

$$\begin{aligned} \langle b_2, b_1 \rangle &= \langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rangle \\ &= [1 \ \ 0] A \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= 0 \end{aligned} \tag{1.41}$$

$$\begin{aligned} \langle b_2, b_2 \rangle &= \langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rangle \\ &= [1 \ \ 0] A \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= 1 \end{aligned} \tag{1.42}$$

So B is an orthonormal basis

Theorem 1.5.7 (Gram-Schmidt process)

Let V be an inner product space and let $\{v_1, \ldots, v_n\}$ be **any basis** of V
Define vectors $b_1, b_2, \ldots, b_n$ in V as follows:

$$\begin{aligned} b_1 &= v_1 \\ b_2 &= v_2 - \frac{\langle v_2, b_1 \rangle}{\|b_1\|^2} b_1 \\ \ldots \\ b_k &= v_k - \frac{\langle v_k, b_1 \rangle}{\|b_1\|^2} b_1 - \frac{\langle v_k, b_2 \rangle}{\|b_2\|^2} b_2 - \ldots - \frac{\langle v_k, b_{k-1} \rangle}{\|b_{k-1}\|^2} b_k \end{aligned} \tag{1.43}$$

for each $k = 2, \ldots, n$. Then
- $\{b_1, \ldots, b_n\}$ is an orthogonal basis of V
- $\operatorname{span}\{b_1, \ldots, b_k\} = \operatorname{span}\{v_1, \ldots, v_k\}$ for each $k = 1, \ldots, n$
- $\left\{\frac{b_1}{\|b_1\|}, \ldots, \frac{b_n}{\|b_n\|}\right\}$ is an orthonormal basis of V